incorporating kernel functions to capture non-linear relationships.

Combining PCA with Clustering

PCA is often used as a preprocessing step before clustering algorithms like K-Means, enhancing their performance on high-dimensional datasets by reducing noise and computational complexity.

PCA is a fundamental tool for dimensionality reduction, enabling the simplification of complex datasets while preserving their variance. By transforming data into a lower-dimensional space, PCA facilitates visualization, enhances computational efficiency, and improves the interpretability of machine learning models. Despite its limitations, PCA remains a cornerstone technique in data analysis, offering powerful insights and paving the way for more advanced analyses.

Real-World Example: Customer Segmentation Using Python

Customer segmentation is one of the most impactful applications of unsupervised learning in marketing and business analytics. It involves dividing customers into distinct groups based on shared characteristics or behaviors, enabling businesses to tailor their marketing strategies, enhance customer satisfaction, and optimize resource allocation. In this example, we demonstrate how K-Means clustering can be applied to segment customers based on their demographic and behavioral data.